

System Blueprint (*a.k.a.* “Design Doc”)

TNPG: koalas

project: Better Talos

Target ship date: 2026-06-01

roster: Alexandru Cimpoiu, Shafin Kazi, Mustafa Abdullah, Jalen Chen

Name	Email	Primary Role	Secondary Role
Alexandru Cimpoiu	alexandruc4@nycstudents.net	Database organizer/parser	Grade prediction algorithm developer
Shafin Kazi	shafink3@nycstudents.net	Javascript	(Secondary) frontend designer
Mustafa Abdullah	mustafaa80@nycstudents.net	API Integration/JS	Middleware manager
Jalen Chen	jalenc60@nycstudents.net	Frontend designer	Secondary database parser

Summary

Better Talos is to help inform Stuy students inform themselves on their classes (e.g. course rigor, course requirements, teacher evaluation, etc) and engage in public forums to discuss all things Stuy.

Problem Being Solved

Talos does not present all of the information that students might need in an easy/meaningful way. Students might have to socialize with others in order to glean information about courses/teachers and better inform themselves on which classes to take or not to take.

Target Users

Who will use this system? - Current Stuyvesant students - Current Stuyvesant administration (guidance counselors, program office)

Why This Project Matters

The project would help students better map out their academic path over the course of their 4 years at Stuyvesant as opposed to the current rudimentary system of blindly picking courses through Talos.

Minimum Viable Product (MVP) Scope

Core Features (Required for Final Submission)

Features that **must** be completed: 1. Online message board for discussion 2. Admin User with moderation abilities 3. Information on each course and teacher 4. Prerequisites for each course

Stretch Features (Only if MVP is Complete)

1. Class Grade predictor
2. Workload difficulty predictor (number of hours hw)

Explicit Non-Goals

Features intentionally excluded: - Real-time collaborative schedule editing - Direct integration/authentication with official DOE or Stuyvesant administrative servers

Technology Stack

Layer	Selected Tool
Backend Framework	Flask
Frontend Framework	tailwind
Database	SQLite
Authentication	Flask sessions
Data viz	d3.js
ORM / DB Library	N/A

Why This Stack Was Chosen

We chose most of these options because they are what we are most comfortable and familiar with. Tailwind offers easy, customizable, and lightweight designing.

Team Ownership Plan

Each member must own meaningful deliverables.

Team Member	Primary Ownership	Secondary Ownership	Specific Deliverables
Alexandru	DB & core algorithms	Data viz	parsing course catalog data, developing the GPA/Workload prediction algorithms, and implementing the d3.js prerequisite tree visualizations
Shafin	Frontend + DOM Manipulation	UI/UX	Building the drag and drop JavaScript logic for the 4 year planner, managing frontend state
'Stafa	Backend Routing & API Endpoints	Middleware + auth	making flask routes, handling Flask session authentication and validating schedule constraints
Jalen	UI/UX	Data parsing	Tailwind + create visual layout

Component map

Site map

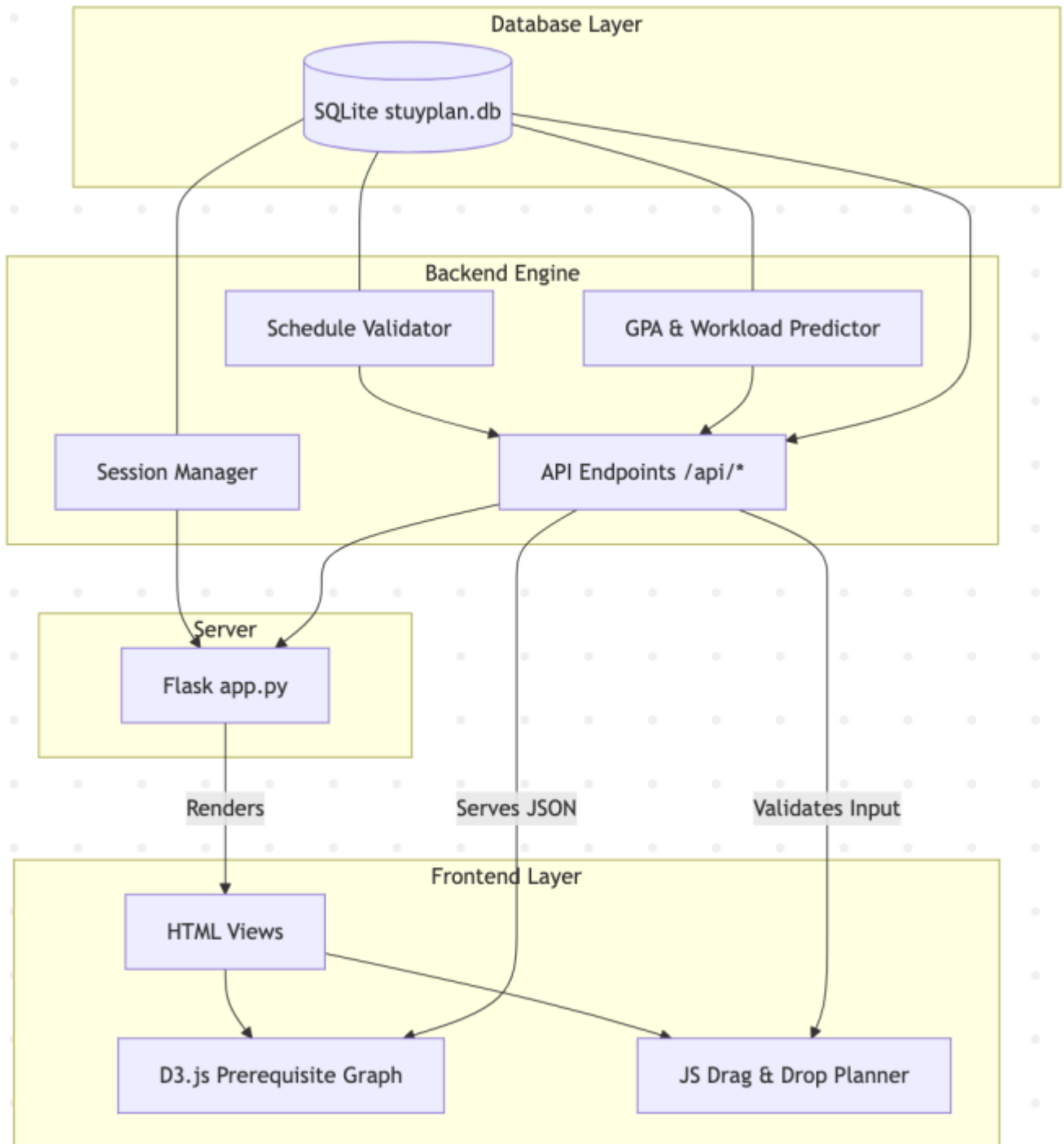


Figure 1: Component Map

Figure 1: Component Map

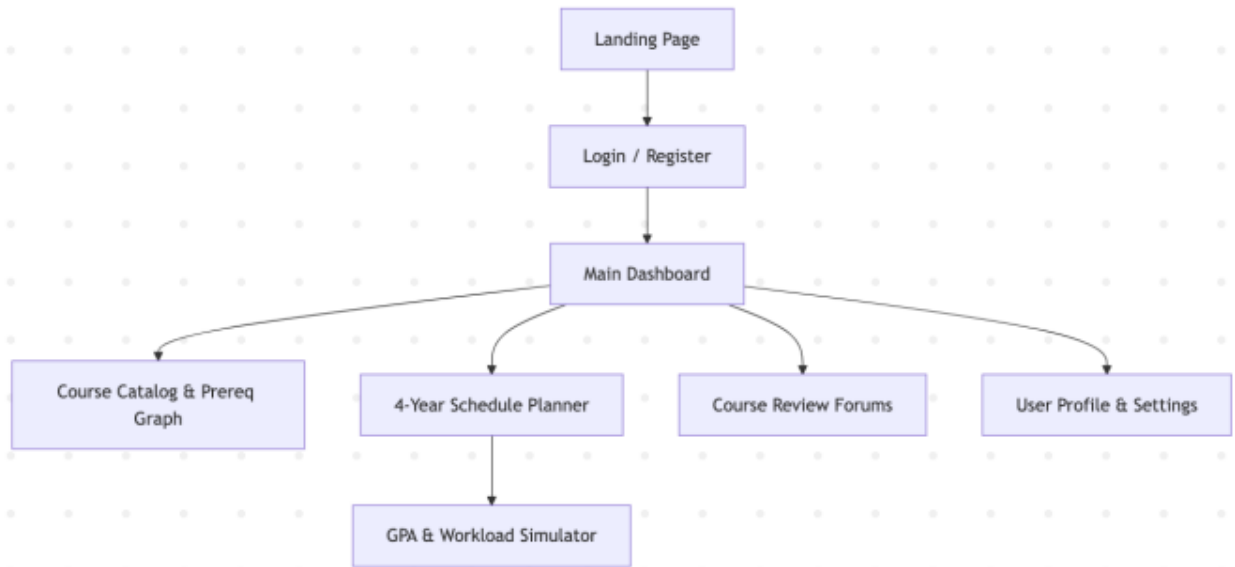


Figure 2: Site Map

Figure 2: Site Map

Key User Stories

eg0

As a **student**, I want to visualize course prerequisites as a graph so that I can see the fastest path to advanced classes without missing required courses.

eg1

As a student, I want to simulate my projected workload and gpa so that I can take classes that are well suited for me without too much burnout.

eg2

As a user, I want to filter course reviews by tags so that I can select classes that align with my personal preferences.

Database Design

Table Name	Fields	Description
Users	id (INTEGER, PK), username (TEXT), password(TEXT), grad_year (INTEGER)	Stores basic user account and auth info
Courses	id (PK, INTEGER), course_code (TEXT), name (TEXT), subject (TEXT), prereqs (TEXT)	All the courses.
Reviews	id (PK, INTEGER), course_code (FK, INTEGER), user_id (FK), difficulty, workload_hours, tags, content	Crowdsourced data for the forum and predictor algorithms

Table Name	Fields	Description
Schedules	id (PK, INTEGER), user_id (FK, INTEGER), course_id (FK, INTEGER), semester_num (INTEGER), year (INTEGER)	Stores a user's planned 4 year schedule

Testing Plan

- Use unit tests to validate that prereqs are working properly.
- Verify prereq data to make sure no dependencies are circular.
- Make sure routes work and fix server errors before pushing to our “prod” environment.

Timeline

Week 1 Goals:

- Visualization of Courses & Prereqs data (proof of concept)
- flask app & sqlite db
- basic frontend created.

Week 2 Goals:

- d3.js graph created using courses & prereqs data
- frontend UI for a drag & drop planner

Week 3 Goals:

- connect drag and drop planner to a backend validator
- integrate the workload/GPA prediction algorithms based on database review aggregates

Internal Deadlines:

- May 18: db fully built with course data & functioning routing
- May 20: functioning prereq site
- May 22: d3 graph created with the prereq database
- May 25: drag & drop schedule builder js made
- May 27: backend fully complete
- May 29: middleware fully completed with full algorithms & page holding student data like classes taken made
- June 1: Finishing touches, bug fixing, whatever crunch is left

Completion Criteria (*a.k.a.* “Definition of ‘Done’ ”)

Project is considered complete when all of the following are true: 1. Meet all MVP requirements 2. Happy and ready to graduate >w<

Open Questions

- How are we going to get data for this?

Appendix

Workload Algorithm: - It will probably be something simple like (Sum of predicted course hours) * User Speed User speed will be calculated by performance above or below average on assignments. Self-reported if no data.